

Improving Quantization-aware Training of Low-Precision Network via Block Replacement on Full-Precision Counterpart

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Abstract—Quantization-aware training (QAT) is a common paradigm for network quantization, in which the training phase incorporates the simulation of the low-precision computation to optimize the quantization parameters in alignment with the task goals. However, direct training of low-precision networks generally faces two obstacles: 1. The low-precision model exhibits limited representation capabilities and cannot directly replicate full-precision calculations, which constitutes a deficiency compared to full-precision alternatives; 2. Non-ideal deviations during gradient propagation are a common consequence of employing pseudo-gradients as approximations in derived quantized functions. In this paper, we propose a general QAT framework for alleviating the aforementioned concerns by permitting the forward and backward processes of the low-precision network to be guided by the full-precision partner during training. In conjunction with the direct training of the quantization model, intermediate mixed-precision models are generated through the block-by-block replacement on the full-precision model and working simultaneously with the low-precision backbone, which enables the integration of quantized low-precision blocks into full-precision networks throughout the training phase. Consequently, each quantized block is capable of: 1. simulating full-precision representation during forward passes; 2. obtaining gradients with improved estimation during backward passes. We demonstrate that the proposed method achieves state-of-the-art results for 4-, 3-, and 2-bit quantization on ImageNet and CIFAR-10. The proposed framework provides a compatible extension for most QAT methods and only requires a concise wrapper for existing codes.

I. INTRODUCTION

Convolutional neural networks (CNNs) have made substantial advancements in image classification [15, 29], semantic segmentation [11, 17], object detection [43, 44], and image restoration [13]. With the large computational complexity of CNNs, there is an increasing demand for strategies to successfully deploy networks on resource-constrained devices with limited processing capability. Several strategies have been developed to train fast and compact neural networks that are both efficient in terms of speed and size, such as manually-designed architectures [7, 20, 47, 55], pruning [10, 32, 35], quantization [1, 8, 14, 22, 33], knowledge distillation [19, 45, 49], and neural architecture search. In this paper, we focus on network quantization, which attempts to transform the weights and/or activations of a full-precision (FP) network into low-precision (LP) approximations so that fixed-point

computations can be conducted without compromising too much accuracy.

Quantization-aware training (QAT) has become one of the most effective methods for achieving low-bit quantization while maintaining relatively high accuracy by simulating low-precision operators during training and optimizing quantizer parameters with task objectives [8, 27, 39]. The main difficulty in directly training a quantized network arises from the discretization stage, in which a discrete quantizer (i.e., scaled round function) converts a normalized value to one of the discrete levels. First, since a discrete quantizer is non-differentiable, the straight-through estimator (STE) [2] is often utilized during backpropagation [5, 8, 25, 31, 41, 53, 59]. Despite recent STE-based methods demonstrating reasonable performance [3, 8, 25, 31, 41], STE may produce an unexpected gradient mismatch issue [31, 52] with distinct forward and backward passes, resulting in an undesirable drop in accuracy [61]. Second, due to the limited representation capability of the discrete quantizer [26], the accuracy of extremely low-bit quantized networks (i.e., 4-, 3-, or 2-bit) is inevitably diminished. In order to enhance the low-bit representation, certain methods are suggested, known as non-uniform quantization [33, 34, 37, 51, 58]. This technique entails modifying the quantization resolution in accordance with the density of the real-valued distribution in order to find suitable correspondences with the activation and weight distributions [33, 51]. However, the increased inference burden presents them with major obstacles.

By consulting full-precision models for guidance as auxiliary supervision, optimal schemes are proposed to partially compensate for the inherent issues of QAT [26, 60, 61] through the incorporation of additional losses into intermediate layers to address the gradient approximation problem and provide regularization. In this regard, the simplest approach commonly employed in QAT is to initialize the low-precision model directly with the weights obtained from the pre-trained full-precision model [3, 8, 33]. Besides, training strategies such as knowledge distillation have been combined to learn a low-precision student network by distilling knowledge from a full-precision teacher [26, 42, 60]; additionally, some strategies use full-precision auxiliary routines for dealing with quantizer noise [4, 61].